Gujarat Technological University

NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY/COMPUTER SCIENCE AND ENG.

ACADEMIC YEAR – 2024-2025

Summer Internship (3170001)

Predicting Customer Response to Telemarketing Campaigns

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Outline

1. Objective

Main Objective: increase the effectiveness of the bank's telemarketing campaign This project will enable the bank to develop a more granular understanding of its customer base, predict customers' response to its telemarketing campaign and establish a target customer profile for future marketing plans.

By analysing customer features, such as demographics and transaction history, the bank will be able to predict customer saving behaviours and identify which type of customers is more likely to make term deposits. The bank can then focus its marketing efforts on those customers. This will not only allow the bank to secure deposits more effectively but also increase customer satisfaction by reducing undesirable advertisements for certain customers.

2. Introduction to Python Programming

Python is widely recognized as a versatile programming language highly valued for its simplicity, clarity, and adaptability, making it a foundational tool in the field of data science.

Its broad applicability spans areas such as statistical analysis, machine learning, and natural language processing, catering to both beginners and experts due to its user-friendly syntax.

Python's extensive standard library offers a wide array of modules and packages tailored to various data-related tasks, including data manipulation, visualization, and machine learning algorithms. Additionally, its support for different programming paradigms, such as procedural, object-oriented, and functional programming, provides data scientists with unmatched flexibility in designing their solutions.

In the realm of data science, Python is complemented by a diverse range of specialized libraries and frameworks, each designed for specific stages of the data analysis process.

For instance, Pandas, NumPy, and Matplotlib excel in data manipulation, numerical computation, and visualization, while scikit-learn and TensorFlow provide comprehensive toolsets for machine learning and deep learning tasks.

3. Role and Responsibilities During Internship

"Predicting Customer Response to Telemarketing Campaigns," here are some roles and responsibilities i have taken on during my internship:

1. Data Collection and Preprocessing

2. Feature Engineering

3. Model Building

4. Model Evaluation

5. Implementation and Deployment

6. Reporting and Documentation

7. Collaboration

8. Continuous Learning

4. Internship Work

- **AIM:** This project will enable the bank to develop a more granular understanding of its customer base, predict customers' response to its telemarketing campaign and establish a target customer profile for future marketing plans.

- By analysing customer features, such as demographics and transaction history, the bank will be able to predict customer saving behaviours and identify which type of customers is more likely to make term deposits. The bank can then focus its marketing efforts on those customers. This will not only allow the bank to secure deposits more effectively but also increase customer satisfaction by reducing undesirable advertisements for certain customers.

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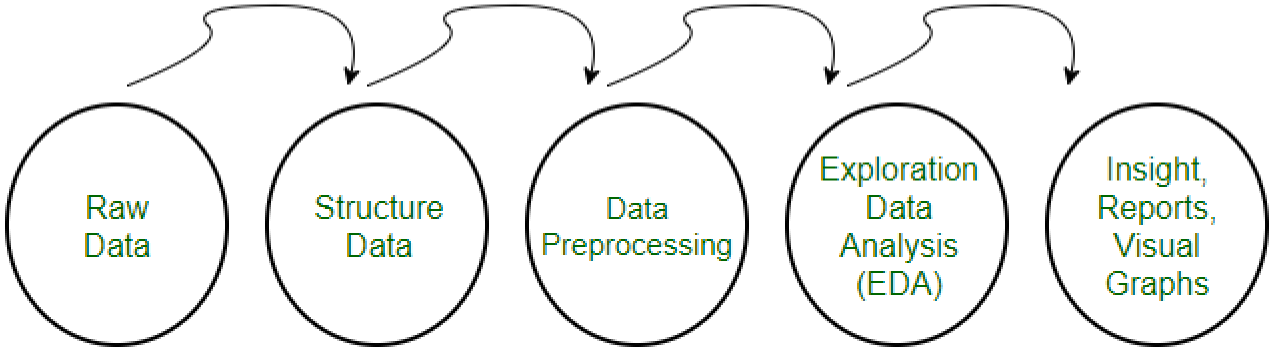
Project Background

Nowadays, marketing spending in the banking industry is massive, meaning that it is essential for banks to optimize marketing strategies and improve effectiveness. Understanding customers’ need leads to more effective marketing plans, smarter product designs and greater customer satisfaction.

Data Cleaning



Load the raw data

This dataset is about the direct phone call marketing campaigns, which aim to promote term deposits among existing customers, by a Portuguese banking institution from May 2008 to November 2010. It is publicly available in the UCI Machine learning Repository, which can be retrieved from http://archive.ics.uci.edu/ml/datasets/Bank+Marketing#.

```python
import pandas as pd
import numpy as np
from pandas import read_csv
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
import os

linkname = 'bank-full.csv'
dataset1 = pd.read_csv(linkname, sep = ';')
```

[1] Python

There are 41,188 observations in this dataset. Each represents an existing customer that the bank reached via phone calls.

- For each observation, the dataset records **16 input variables** that stand for both qualitative and quantitative attributes of the customer, such as age, job, housing and personal loan status, account balance, and the number of contacts.
- There is **a single binary output variable** that denotes "yes" or "no" revealing the outcomes of the phone calls.

```python
# View the first 5 rows in the dataset
dataset1.head()
```

[2] Python

	age	job	marital	education	default	balance	housing	loan	contact	day	month	duration	campaign	pdays	previous	poutcome	y
0	58	management	married	tertiary	no	2143	yes	no	unknown	5	may	261	1	-1	0	unknown	no
1	44	technician	single	secondary	no	29	yes	no	unknown	5	may	151	1	-1	0	unknown	no
2	33	entrepreneur	married	secondary	no	2	yes	yes	unknown	5	may	76	1	-1	0	unknown	no
3	47	blue-collar	married	unknown	no	1506	yes	no	unknown	5	may	92	1	-1	0	unknown	no
4	33	unknown	single	unknown	no	1	no	no	unknown	5	may	198	1	-1	0	unknown	no

Clean the dataset

Clean the dataset

2.1 Deal with missing data

There is no missing value in this dataset. Nevertheless, there are values like “unknown”, “others”, which are helpless just like missing values. Thus, these ambiguous values are removed from the dataset.

```python
# Step 1: Delete the rows which column 'poutcome' contains 'other'
condition = dataset1.poutcome == 'other'
dataset2 = dataset1.drop(dataset1[condition].index, axis = 0, inplace = False)
```

[3] Python

```python
# Step 2: Replace 'unknown' in job and education with 'other'
dataset2[['job','education']] = dataset2[['job','education']].replace(['unknown'],'other')
```

[4] Python

2.2 Drop outliers in the column 'balance'

In order to capture the general trend in the dataset, outliers in the column “balance” are dropped. Outliers are defined as the values which are more than three standard deviations away from the mean. In sum, 2556 rows of data were removed.

```python
from scipy.stats import zscore

dataset2[['balance']].mean()
dataset2[['balance']].mean()

dataset2['balance_outliers'] = dataset2['balance']
```

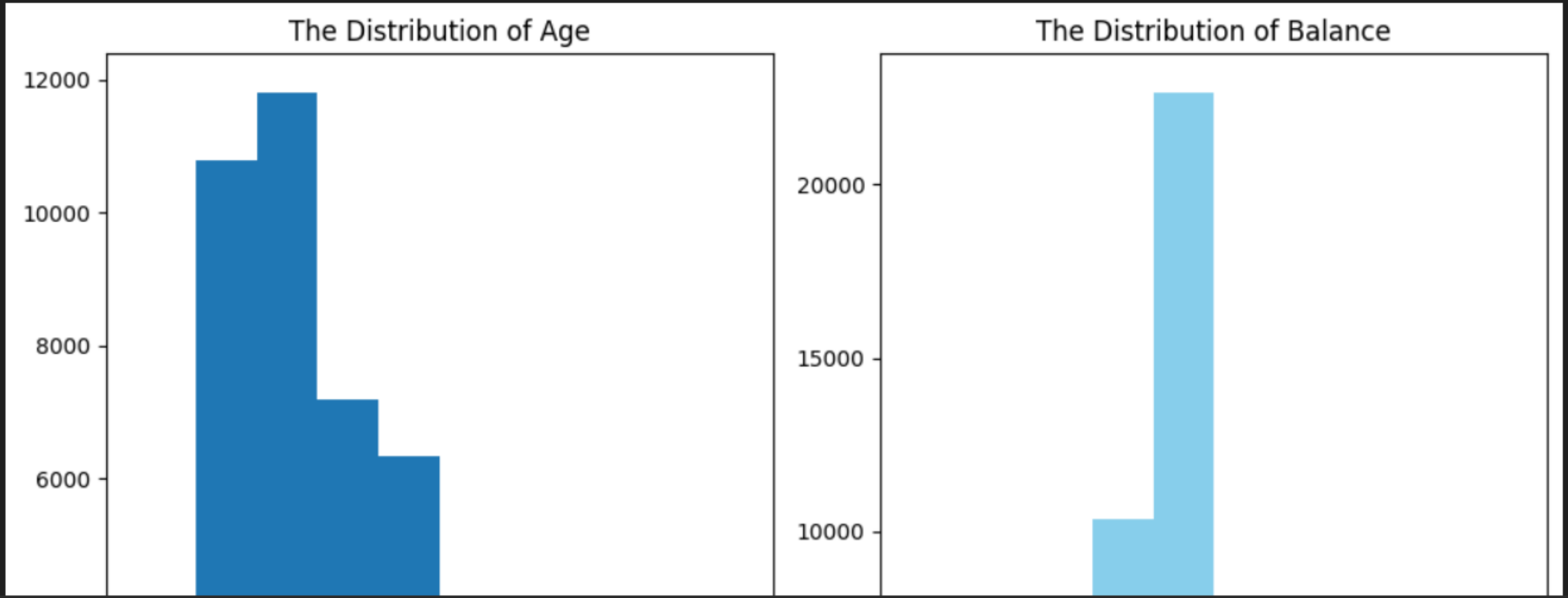
Exploratory data analysis

To obtain a better understanding of the dataset, the distribution of key variables and the relationships among them were plotted.

3.1 Visualize the distribution of 'age' and 'balance'

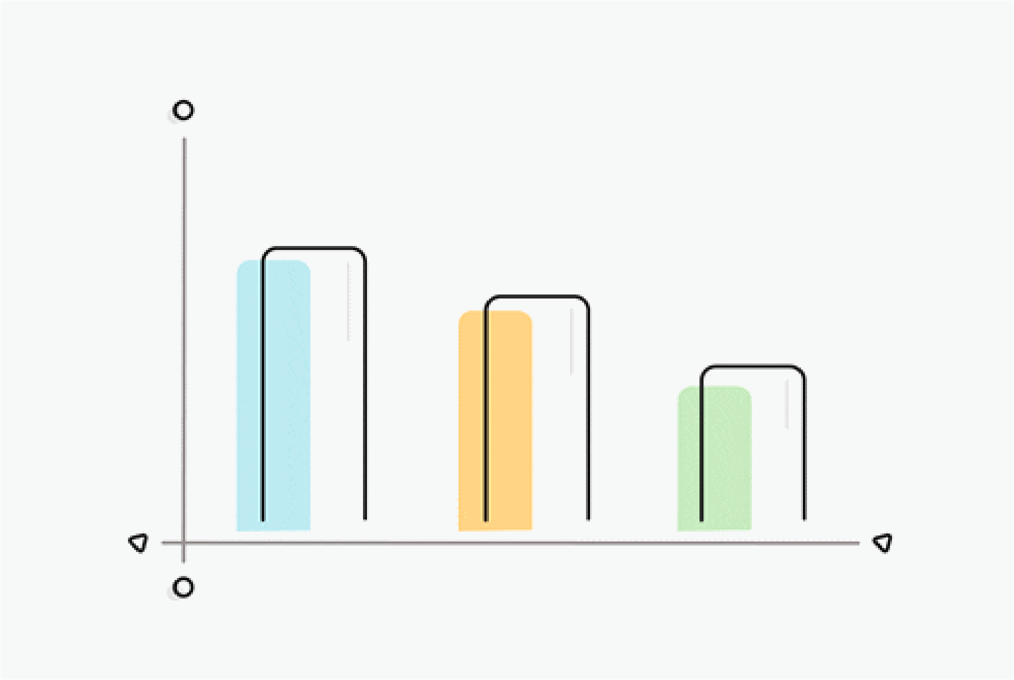
```
dist_age_balance = plt.figure(figsize = (10,6))

ra1 = dist_age_balance.add_subplot(1,2,1)
ra2 = dist_age_balance.add_subplot(1,2,2)

ra1.hist(dataset7['age'])
ra1.set_title('The Distribution of Age')

ra2.hist(dataset7['balance'], color = 'skyblue')
ra2.set_title('The Distribution of Balance')

plt.tight_layout()
plt.show()
```

[13] Python



Data Visualization

With a sound knowledge of the distribution of key variables, further analysis of each customer characteristic can be carried out to investigate its influence on the subscription rate.



4.1 Visualize the subscription and contact rate by age

```python
lst = [dataset7]
for column in lst:
    column.loc[column["age"] < 30,  'age_group'] = 20
    column.loc[(column["age"] >= 30) & (column["age"] <= 39), 'age_group'] = 30
    column.loc[(column["age"] >= 40) & (column["age"] <= 49), 'age_group'] = 40
    column.loc[(column["age"] >= 50) & (column["age"] <= 59), 'age_group'] = 50
    column.loc[column["age"] >= 60, 'age_group'] = 60
```

[19] Python

```python
count_age_response_pct = pd.crosstab(dataset7['response'],dataset7['age_group']).apply(lambda x: x/x.sum() * 100)
count_age_response_pct = count_age_response_pct.transpose()
```

[20] Python

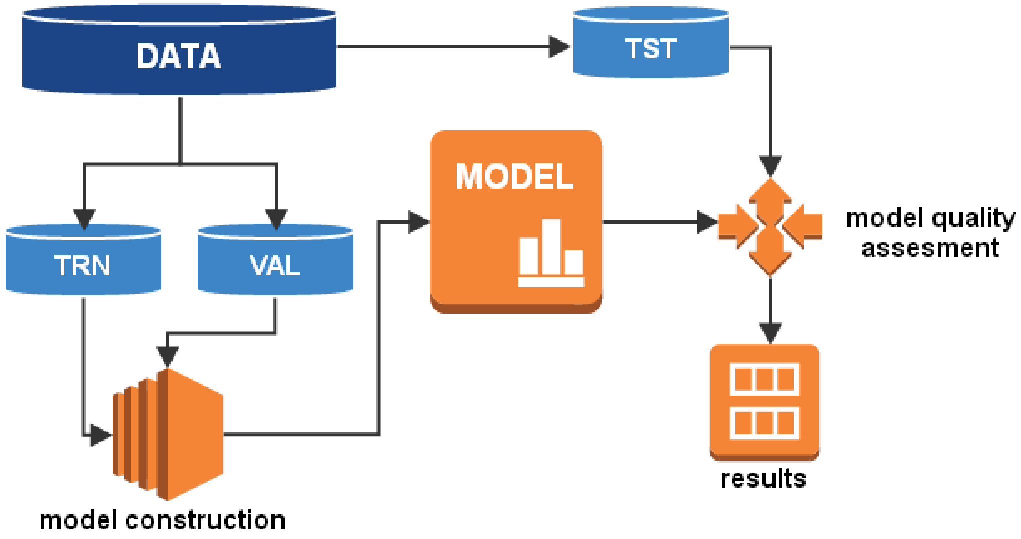
```python
age = pd.DataFrame(dataset7['age_group'].value_counts())
age.rename(columns={'count': 'age_group'}, inplace=True)  # Rename the 'count' column to 'age_group'
age['% Contacted'] = age['age_group']*100/age['age_group'].sum()
age['% Subscription'] = count_age_response_pct['yes']
age.drop('age_group',axis = 1,inplace = True)

age['age'] = [30,40,50,20,60]
age = age.sort_values('age',ascending = True)
```

[21] Python

```python
plot_age = age[['% Subscription','% Contacted']].plot(kind = 'bar',
                                                      figsize=(8,6), color = ('green','red'))
plt.xlabel('Age Group')
plt.ylabel('Subscription Rate')
plt.xticks(np.arange(5), ('<30', '30-39', '40-49', '50-59', '60+'),rotation = 'horizontal')
plt.title('Subscription vs. Contact Rate by Age')
plt.show()
```

Machine Learning: Classification

The main objective of this project is to identify the most responsive customers before the marketing campaign so that the bank will be able to efficiently reach out to them, saving time and marketing resources. To achieve this objective, classification algorithms will be employed. By analysing customer statistics, a classification model will be built to classify all clients into two groups: "yes" to term deposits and "no" to term deposits.



Prepare Data for Classification

5.1 Select variables relevant to customers

Only the most relevant customer information is considered, which includes job title, education, age, balance, default record, housing record and loan record. Other information, such as ‘the number of contacts performed before this campaign’, is omitted because it is not directly related to customers themselves.

```python
dataset.drop(['marital'],axis=1, inplace=True)
dataset1 = dataset.iloc[:, 0:7]
```

[36] Python

5.2 Tranform categorical data into dummy variables

Since machine learning algorithms only take numerical values, all five categorical variables (job, education, default, housing and loan) are transformed into dummy variables.

Dummy variables were used instead of continuous integers because these categorical variables are not ordinal. They simply represent different types rather than levels, so dummy variables are ideal to distinguish the effect of different categories.

```python
dataset2 = pd.get_dummies(dataset1, columns = ['job'])
dataset2 = pd.get_dummies(dataset2, columns = ['education'])
dataset2['housing'] = dataset2['housing'].map({'yes': 1, 'no': 0})
dataset2['default'] = dataset2['default'].map({'yes': 1, 'no': 0})
dataset2['loan'] = dataset2['loan'].map({'yes': 1, 'no': 0})
dataset_response = pd.DataFrame(dataset['response_binary'])
dataset2 = pd.merge(dataset2, dataset_response, left_index = True, right_index = True)
```

[37] Python

Machine Learning: Regression

Regression analysis is carried out to complement the classification result and help the bank better predict campaign outcome based on customer statistics.

Since the duration of a phone call is positively correlated with the campaign outcome, it can serve as another indicator of the possibility of subscription. In this part, regression algorithms will be used to estimate the duration of a phone call, helping the bank better predict subscription rate.

Prepare Data for Regression

```
dataset4 = dataset2.drop(['response_binary'],axis = 1)
dataset4['duration'] = dataset['duration']
```

[51] Python

6.1 Feature selection

The values of the first 19 columns, which contain customer statistics, are selected as features while the value of the last column, 'duration', is set as target.

```
array = dataset4.values
X = array[:,0:20]
Y = array[:,20]
```

[52] Python

6.2 Train/ test split

```
test_size= 0.20
seed = 10
X_train, X_test, Y_train, Y_test= train_test_split(X, Y, test_size=test_size, random_state=seed)
```

[53] Python

Compare regression algorithms

Six different regression algorithms (Linear Regression, Lasso, Ridge, ElasticNet, K Neighbors and Decision Tree) are run on the dataset and the best-performing one will be used to build the

5. Conclusion

According to previous analysis, a target customer profile can be established. The most responsive customers possess these features:

* Feature 1: age < 30 or age > 60

* Feature 2: students or retired people

* Feature 3: a balance of more than 5000 euros

By applying logistic and ridge regression algorithms, classification and estimation model were successfully built. With these two models, the bank will be able to predict a customer's response to its telemarketing campaign before calling this customer. In this way, the bank can allocate more marketing efforts to the clients who are classified as highly likely to accept term deposits, and call less to those who are unlikely to make term deposits.

In addition, predicting duration before calling and adjusting marketing plan benefit both the bank and its clients. On the one hand, it will increase the efficiency of the bank’s telemarketing campaign, saving time and efforts. On the other hand, it prevents some clients from receiving undesirable advertisements, raising customer satisfaction. With the aid of logistic and ridge regression models, the bank can enter a virtuous cycle of effective marketing, more investments and happier customers.

6. Future Enhancement

Recommendations

1. More appropriate timing

When implementing a marketing strategy, external factors, such as the time of calling, should also be carefully considered. The previous analysis points out that March, September, October and December had the highest success rates. Nevertheless, more data should be collected and analyzed to make sure that this seasonal effect is constant over time. If the trend has the potential to continue in the future, the bank should consider initiating its telemarketing campaign in fall and spring.

2. Smarter marketing design

By targeting the right customers, the bank will have more and more positive responses, and the classification algorithms would ultimately eliminate the imbalance in the original dataset. Hence, more accurate information will be presented to the bank for improving the subscriptions. Meanwhile, to increase the likelihood of subscription, the bank should re-evaluate the content and design of its current campaign, making it more appealing to its target customers.

3. Better services provision

With a more granular understanding of its customer base, the bank has the ability to provide better banking services.

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Completion Certificate

This is to confirm that **Mr/Ms. Gopeshkumar harsukh rabadiya (211430116017)**, **New L J institute of Engineering & Technology** (B.E.-IT) has successfully completed an internship Programme in **Data Science & Machine Learning** from **29th Jun-2024 to 12th Jul-2024**, in our company.

Intern worked really hard and was tremendously driven. Intern put in a lot of effort and performed a great job on the duties at hand.

We hope all of the future undertakings of the intern would be a tremendous success.

Su-kEm Tech Lab
Ahemdabad

Regards,
Vrunda Hingarajia
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THANK
YOU!